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Research article



Integrated aerobic-anaerobic digestion of highly solids-loaded corn stover and swine manure under dynamic aeration: Temperature rise, physicochemical characteristics, and methane production

Lili Wang [1], Zicong Wang [1], Zhongjiang Wang [*], Jingke Zheng

College of Engineering, Northeast Agricultural University, Harbin, 150030, China

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ABSTRACT

This research aimed to design an integrated aerobic-anaerobic reactor with dynamic aeration that was automatically regulated based on real-time oxygen concentration and investigate the aerobic pretreatment and subsequent dry co-anaerobic digestion (co-AD) characteristics of highly solids-loaded corn stover and swine manure in terms of temperature rise, physiochemical characteristics, and methane production. The high-temperature feedstocks from the aerobic pretreatment phase rapidly entered the AD phase without transportation and effectively improved the start-up and methane production of the co-AD. Oxygen concentration range, aeration rate, and pretreatment time affected the cumulative aeration time, temperature rise, and organic matter removal interactively during aerobic pretreatment, and a low aeration rate was relatively preferable. Although the lignocellulose removal increased with the increase in pretreatment duration, the maximal lignin elimination efficiency only reached 1.30%. The inoculum injection in the transition phase from aerobic pretreatment to co-AD and the leachate reflux during co-AD were also critical for producing methane steadily apart from aerobic pretreatment. The cold air weakened the temperature rise of aerobic pretreatment, and the low-temperature leachate reduced the methane production in the co-AD process. An oxygen concentration range of 13%–17%, aeration rate of 0.10 $m^3/(min{\cdot}m^3)$, pretreatment time of 84 h, inoculum loading of 40%, leachate refluxing thrice per day, and double-layer inoculation were optimum for improving the integrated aerobic-anaerobic digestion system's ability to resist low temperatures and achieving high methane production. The maximal cumulative and volatile solids (VS) methane yields of corn stover and swine manure reached 444.58 L and 266.30 L/kg VS.

1. Introduction

As a primary grain-producing and livestock husbandry country, the annual outputs of lignocellulosic biomass and livestock manure reach over 0.7 and 3.05 billion tons in China, with corn stover and swine manure representing about a third of each (Dong et al., 2023; Zhang et al., 2024; Zhou et al., 2024). Generally, feces and wastewater are collected separately in most Chinese swine breeding farms to save water resources and reduce wastewater production. Therefore, dry anaerobic digestion (AD) featuring high solids loading (generally 15%–35%, *w/w*), low reactor scale cost, and high methane productivity (Karrabi et al., 2023) is an environmentally friendly approach for treating corn stover and swine manure, as well as high-carbon corn stover and high-nitrogen swine manure can balance the carbon-to-nitrogen (C/N) ratio during their co-AD and swine manure itself contains lots of microflora that can quickly adapt to the anaerobic environment and generate methane (Arias et al., 2020). However, dry AD without pretreatment is prone to acid and ammonia inhibition due to the compact lignocellulosic structure of corn stover and the high presence of ammonium in swine manure (Song et al., 2023).

Aerobic pretreatment mainly includes aerobic or microaerobic hydrolysis pretreatment of lignocellulosic biomass under low total solids (TS) loading (≤10%, *w/w*) (Chu et al., 2021; Huang et al., 2016), air stripping aiming at reducing ammonium levels of swine wastewater, digestate et al. before or after anaerobic digestion (Folino et al., 2020), and aerobic composting of mixed organic feedstocks containing

* Corresponding author.
E-mail address: neauwzj@neau.edu.cn (Z. Wang).
[1] The authors contributed equally to this work.

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lignocellulosic biomass, livestock manure et al. under high TS loading (commonly 30%–50%, *w/w*) (Guo et al., 2012; Labatut et al., 2011). Aerobic pretreatment under high solids loading could facilitate an integrated operation of aerobic and dry-anaerobic equipment, where both aerobic and anaerobic processes could be in a single reactor instead of the traditional transportation of pretreated feedstocks from the pretreatment reactor to the anaerobic reactor. Multiple reactors could be operated independently in parallel to match the volume of waste treated at different scales of small, medium, and large, thereby avoiding heat and power losses during feedstock transportation, increasing the flexibility of equipment operation, and reducing the running cost. Meanwhile, aerobic pretreatment can break down the lignocellulosic structure of straw to a certain extent and reduce the ammonium content of feedstocks, which is favorable for following dry AD. Several documents expounded that aerobic pretreatment under high solids loading was conducive to methane production from municipal solid waste and food waste. For example, Fdez-Güelfo et al. (2011) pretreated municipal solid waste and increased the methane yield by 73%. Angeriz-Campoy et al. (2023) found that aerobic pretreatment effectively improved the hydrolysis and acid production stages during the co-AD of municipal solid waste and food waste. Ali et al. (2018) reported that aerobic pretreatment before an anaerobic landfill of municipal solid waste could effectively reduce the lag phase of methane production. However, the aerobic pretreatment and subsequent dry co-AD characteristics of lignocellulosic biomass and livestock manure under high solids loading, especially the integrated operation of aerobic-anaerobic digestion in the same reactor, remain unclear.

In addition, aerobic pretreatment under high solids loading can accumulate a large amount of heat and increase the temperature of feedstocks due to the metabolic reproduction of thermophilic microorganisms, saving power consumption for heating. Meanwhile, some facultative hydrolysis and acid-producing bacteria favorable for AD can enrich during aerobic pretreatment (Lin et al., 2018). However, too much pretreatment time or excessive ventilation can cause undue loss of energy and organic matter, and too little treatment time or deficient ventilation can weaken the pretreatment effect. Dynamic aeration, which relies on real-time oxygen concentration to control aeration, is mainly employed in wastewater treatment (Herrmann-Heber et al., 2020). For aerobic composting, most of the attention focuses on monitoring the oxygen concentrations in the exhaust gas at different stages of the composting process to study the relationship between microbial activity and oxygen concentrations (Nie, 2000; Puyuelo et al., 2010). It would not only provide a more suitable oxygen amount to ensure the metabolism and reproduction of microorganisms, accelerate warming, and improve pretreatment efficiency but also further reduce energy consumption and heat loss compared to traditional aeration if dynamic aeration is employed in the aerobic pretreatment procedure (Zeng et al., 2022). Therefore, dynamic aeration is promising for enhancing aerobic pretreatment and subsequent dry AD. In addition, methanogens-enriched inoculum can accelerate the start of AD, avoid rancidity, and increase the decomposition rate of feedstocks. However, too much inoculum would affect the efficiency of volumetric methane production and even inhibit the AD process. The appropriate amount of inoculum also varies greatly depending on the feedstocks, moisture content, C/N ratio, and so on (Axelle et al., 2016). The leachate reflux can reduce the accumulation of volatile fatty acids (VFAs) and ammonia during dry AD due to the mixing difficulty under high solids loading and improve mass and heat transfer efficiencies. However, excessive flushing of leachate would destabilize the microbial communities, affecting methane production (Song et al., 2023). Accordingly, the inoculum injection after aerobic pretreatment and before dry co-AD of corn stover and swine manure and the leachate reflux during their co-AD are also critical to ensure stable operation of the integrated aerobic-anaerobic process. Therefore, exploring the aerobic pretreatment characteristics and control parameters of corn stover and swine manure with high solids loading under dynamic aeration and the methanogenic properties of subsequent dry co-AD are valuable.

This study aims to design a set of aerobic-anaerobic integrated devices with a dynamic aeration system automatically regulated based on the oxygen concentrations and to investigate the feasibility of fulfilling the aerobic and dry anaerobic processes in a single reactor. First, the aerobic pretreatment and subsequent co-AD characteristics of corn stover and swine manure with high solids loading under dynamic aeration were studied systematically in terms of temperature rise, physicochemical characteristics, and methane production. Second, the effects of the inoculum addition between aerobic pretreatment and dry co-AD and the leachate reflux in the dry co-AD phase on methane production were investigated to realize the rapidly non-intermittent conversion from aerobic pretreatment to dry co-AD and the stable operation of the dry co-AD process. The information obtained will help to fill the gaps in the literature on aerobic pretreatment and methane production from highly solids-loaded complex biomass and accelerate the widespread application of dynamic aeration and scale development of integrated aerobic-anaerobic digestion.

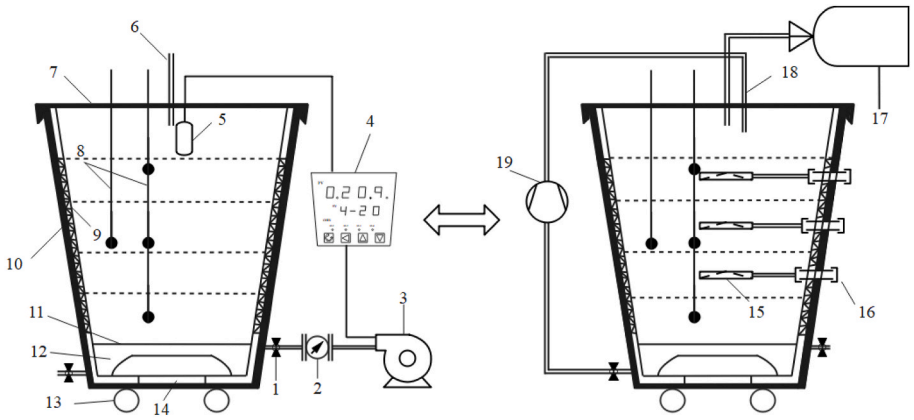
2. Materials and methods

2.1. Feedstocks and inoculum

Air-dried corn stover taken from the corn field of Northeast Agricultural University in China was crushed into 3–5 mm particle size by a pulverizer (TQ-2500Y, Yongkang Tianqishengshi Instrument Co., Ltd., Zhejiang, China) and stored at 20 ± 1.5 °C before use. The TS, VS, total carbon (TC), total nitrogen (TN), cellulose, hemicellulose, and lignin contents of corn stover were 91.85 ± 0.53%, 84.61 ± 0.41%, 44.18 ±0.018%, 0.68 ± 0.02%, 39.35 ± 0.35%, 29.21 ± 0.28% and 12.04 ± 0.18%, respectively. Swine manure collected from the Guangming Village breeding base in China was kept at −20 °C until use. The inoculum for co-AD was the biogas slurry of swine manure digested in the laboratory. The swine manure was anaerobically digested in a 10 L jar at 35 °C until the methane production stopped to obtain digestate. The slurry obtained after removing solids from the digestate was used as inoculum to avoid clogging in the inoculum addition process. The TS, VS, TC, TN, cellulose, hemicellulose, and lignin contents and the pH value of swine manure were 27.83 ± 0.65%, 20.41 ± 0.28%, 25.34 ± 0.49%, 1.86 ± 0.13%, 18.17 ± 0.16%, 21.51 ± 0.32%, 7.24 ± 0.39%, and 6.79 ± 0.41, respectively. The TS, VS, and TN contents and the pH value of inoculum were 3.28 ± 0.07%, 1.34 ± 0.11%, 5.89 ± 0.26%, and 7.23 ± 0.03, respectively.

2.2. Aerobic-anaerobic integrated devices with dynamic aeration system

An integrated aerobic-anaerobic reactor is a self-designed cylindrical polypropylene barrel with a total volume of 80 L, an upper inner diameter of 550 mm, a lower inner diameter of 400 mm, a height of 750 mm, and a thickness of 2.5 mm, as shown in Fig. 1. The entire outer wall of the reactor is insulated using a polyurethane blanket with a thickness of 20 mm (part 10 in Fig. 1). A porous baffle (part 11 in Fig. 1) with a diameter of 450 mm (aperture of 5 mm) was placed on a support frame of the reactor bottom to hold the feedstocks and separate the leachate generated during aerobic pretreatment and co-AD. The aeration holes of 10 mm diameter opened on the upper cover and bottom of the reactor, respectively, could ensure that the air through the airflow meter and air blower (parts 2 and 3 in Fig. 1) was injected from the bottom vent and discharged from the upper vent after forced aeration. Three inlets evenly opened on the side wall of the reactor realized inoculum injection at the upper, middle, and lower layer positions on the central axis of the reactor. Four temperature measurement points set at the upper, middle, lower, and flank positions inside the reactor corresponded to 450–530, 300–380, and 150–230 mm from the bottom of the reactor, and 75–80 mm from the side wall (TEM_{upper}, TEM_{middle}, TEM_{lower}, and TEM_{flank}), respectively. An electrochemical zirconia sensor probe (part 5 in Fig. 1)



1. Ball valve; 2. Flowmeter; 3. Air blower; 4. Oxygen concentration controller; 5. Oxygen sensor; 6. Exhaust outlet pipe; 7. Roof cap; 8. Temperature sensor; 9. Heating belt; 10. Insulation layer; 11. Partition plate; 12. Leachate collection chamber; 13. Universal wheel; 14. Roller bracket; 15. Inoculum injection position; 16. Inoculation port; 17. Biogas collecting bag; 18. Leachate reflux pipe; 19. Pump.

Fig. 1. Aerobic-anaerobic integrated reactor with dynamic aeration system.

installed at the exhaust port constantly measured the oxygen concentrations in the exhaust gas. The oxygen concentration control system started the air blower to aerate when the oxygen sensor detected that the oxygen concentration was lower than the lower limit threshold value, and then closed the air blower and stopped ventilating when the oxygen concentration was higher than the upper limit threshold value, which realized the dynamic aeration based on the real-timely monitored oxygen concentration in the exhaust gas. An electric energy meter connected in series on the line recorded the power consumption during the aeration. Inoculum was added to the pretreated feedstocks from the inoculation ports and the inoculum injection positions (parts 15 and 16 in Fig. 1) after turning off the ventilation system at the end of aerobic pretreatment, and then started the warming system (part 9 in Fig. 1) to begin the AD phase and the collected leachate was refluxed into the feedstocks by a pump (part 19 in Fig. 1) from the roof cap of the reactor daily during dry co-AD.

2.3. Aerobic pretreatment and dry anaerobic digestion

The respective additional amount of corn stover and swine manure directly determine the C/N ratio of mixed feedstocks, and the C/N ratio and ventilation conditions affect the warming in the aerobic pretreatment process (Kong et al., 2024), which in turn influence the subsequent co-AD. Furthermore, the threshold values of the oxygen concentration range and the aeration rate for dynamic aeration were crucial and could interactively affect the pretreatment effect with the pretreatment duration. Nie (2000) and Puyuelo et al. (2010) proposed that the microorganisms were active under oxygen concentrations of 14%–17% and 15%–20% during aerobic composting, respectively. Huang et al. (2019) reported that the metabolic activities of aerobic microorganisms almost stopped when the oxygen concentration was less than 10%. The closer the set upper threshold is to the oxygen concentration of 21% in the ambient air, the longer the aeration process might run, and the closer the set lower threshold is to 10%, the more likely the microorganisms might be in anaerobic conditions (Wang et al., 2024). In addition, Wang et al. (2022b)Wang et al., 2022a,b, achieved an optimal ventilation rate of 0.20 $m^3/(min{\cdot}m^3)$ for aerobic composting, while Xu et al. (2018) recommended a range of ventilation rate [0.05–0.20 $m^3/(min{\cdot}m^3)$] for composting. Therefore, this study first explored the critical control parameters and pretreatment characteristics of the aerobic process and then studied the main factors for the rapid start and stable run of the dry co-AD, namely (1) Aerobic pretreatment experiment I: the effects of the initial C/N ratios of feedstocks on the temperature rise and lignin removal during aerobic pretreatment were first studied. The routinely suitable C/N ratios for aerobic composting and dry AD are 25–35 and 20–30, respectively, with a higher C/N ratio commonly favoring the warming of feedstocks (Kong et al., 2024). Therefore, three C/N ratios of 25, 30, and 35 (CN_1, CN_2, and CN_3) corresponding to the mass ratios of corn stover to swine manure of 0.8, 1.3, and 2 (*w/w*) and three aeration rates of 0.10, 0.15, and 0.20 $m^3/(min{\cdot}m^3)$ (R_1, R_2, and R_3) were studied. The oxygen concentration range, aerobic pretreatment duration, initial moisture content (MC), and mean room temperature were 13%–17%, 120 h, 60%, and 19.3 ± 2.4 °C, respectively. (2) Aerobic pretreatment experiment II: the temperature rise and physicochemical characteristics of corn stover and swine manure during aerobic pretreatment under dynamic aeration were studied systematically. Three oxygen concentration ranges of 9%–13%, 13%–17%, and 17–20% (Aer_1, Aer_2, and Aer_3), three aeration rates of 0.10, 0.15, and 0.20 $m^3/(min{\cdot}m^3)$ (R_1, R_2, and R_3), and three aerobic pretreatment durations of 48, 84, and 120 h (T_1, T_2, and T_3) were investigated based on three-factor and three-level orthogonal experiment [$L_9(3^3)$], as shown in Table 1. The initial MC and mean room temperature were 60% and 20.4 ± 2.1 °C, respectively. (3) Dry co-AD experiment III: The effects of the aerobic pretreatment under dynamic aeration on the co-AD characteristics of corn stover and swine manure were studied. Specifically, the pretreated corn stover and swine manure after a set pretreatment time was inoculated with an inoculum loading of 30% (*w/w*, dry basis) and injected into the two positions of upper and middle layers, adjusted to a TS content of 20% with tap water, and then turned off the ventilation system and started the co-AD process. The constant temperature and duration of co-AD were 35 °C and 25 days, respectively. The leachate collected from the bottom of the co-AD reactor was backflowed twice daily. The corn stover and swine manure without aerobic pretreatment were a control. (4) Dry co-AD experiment IV: based on the optimum pretreated results from (1) to (3), the pretreatment time (A_1: 72 h, A_2: 84 h, A_3: 96 h), inoculum loading (B_1: 20%, B_2: 30%, B_3: 40%; *w/w*, dry basis), leachate reflux repetitions per day (C_1: Once, C_2: Twice, C_3: Thrice), and the number of inoculum layers (D_1: single-layer, D_2: double-layer, and D_3: triple-layer corresponding to the injection postions of the upper layer, upper-middle layers, and upper-middle-lower layers) on the methanogenic characteristics of pretreated corn stover and swine manure were further studied. The mean room temperatures were 18.3 ± 1.7, 17.5 ± 2.3, 17.3 ± 2.4, and

Table 1
$L_9(3^3)$ three-factor and three-level uniform experimental plan.

NO.	Oxygen concentration range/Aer (%)	Aeration rate/R [m^3/(min·m^3)]	Pretreatment time/T (h)
1 ($Aer_1R_1T_1$)	1 (9–13, Aer_1)	1 (0.10, R_1)	1 (48, T_1)
2 ($Aer_1R_2T_2$)	1 (9–13, Aer_1)	2 (0.15, R_2)	2 (84, T_2)
3 ($Aer_1R_3T_3$)	1 (9–13, Aer_1)	3 (0.20, R_3)	3 (120, T_3)
4 ($Aer_2R_1T_2$)	2 (13–17, Aer_2)	1 (0.10, R_1)	2 (84, T_2)
5 ($Aer_2R_2T_3$)	2 (13–17, Aer_2)	2 (0.15, R_2)	3 (120, T_3)
6 ($Aer_2R_3T_1$)	2 (13–17, Aer_2)	3 (0.20, R_3)	1 (48, T_1)
7 ($Aer_3R_1T_3$)	3 (17–20, Aer_3)	1 (0.10, R_1)	3 (120, T_3)
8 ($Aer_3R_2T_1$)	3 (17–20, Aer_3)	2 (0.15, R_2)	1 (48, T_1)
9 ($Aer_3R_3T_2$)	3 (17–20, Aer_3)	3 (0.20, R_3)	2 (84, T_2)

17.1 ± 2.1 °C, respectively, for the aerobic pretreatment experiments of A, B, C, and D groups. 25 g of substrates collected from each temperature measurement point were uniformly mixed and used for physicochemical analysis.

2.4. Analytical methods

The TS, VS, and organic matter (OM) contents were measured by differential gravimetric method (Wang et al., 2020). The pH value and

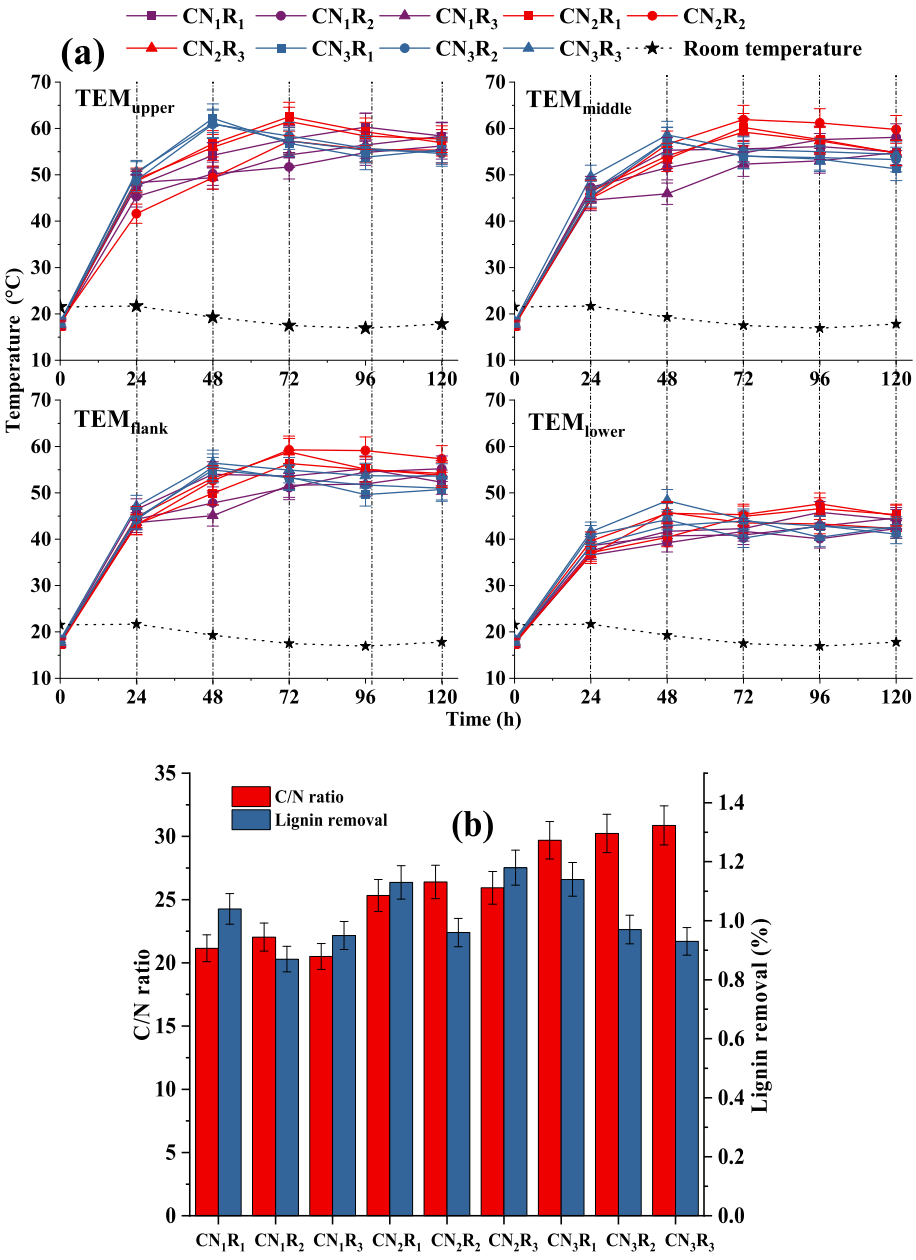


Fig. 2. The effects of the C/N ratio on aerobic pretreatment of corn stover and swine manure under different aeration rates. (a) Temperature rise. (b) Final C/N ratio and lignin removal.

ammonia nitrogen content of leachate were determined by a pH meter (HS-25, Shanghai INESA Scientific Instrument Co., Ltd., Shanghai, China) and an automatic Kjeldahl nitrogen analyzer (K9860, Shandong Hanon Scientific Instrument Co., Ltd., Shandong, China). The TC and TN contents were measured by an elemental analyzer (EA 3000, LEEMAN Technologies Co., Ltd., Beijing, China). The cellulose, hemicellulose, and lignin contents were measured using a fiber analyzer (Ankom 200i, Anke Borui Technologies Co., Ltd., Beijing, China). The compositions of biogas and VFAs were analyzed using a chromatograph (GC-6890 N, Agilent Inc., Santa Clara, CA, USA). All data were reported as the average of duplicates, and statistical analyses were performed with SPSS 22.0.

3. Results and discussion

3.1. Temperature rise and lignin removal characteristics of corn stover and swine manure during aerobic pretreatment under different carbon-to-nitrogen ratios and aeration rates

As shown in Fig. 2 (a), the temperatures under different C/N ratios and aeration rates showed a rapid increase followed by a slight decrease. The upper-layer and middle-layer temperatures were holistically higher than the lower-layer temperatures, and the flank temperatures were similar to the middle-layer temperatures. The lower-layer feedstocks first contacted the incoming air, resulting in more heat dissipation than other layers (Li et al., 2023). Meanwhile, the leachate gradually seeped from the upper-layer and middle-layer feedstocks and gathered into the lower-layer feedstocks, inhibiting the microorganisms because of poor permeability and oxygen diffusion (Guo et al., 2012). Therefore, the lower-layer temperature was the lowest compared to other layers. Moreover, the temperatures under CN_3 in the first 48 h were slightly higher than those under CN_1 and CN_2, while the temperatures under CN_2 from 72 h were higher than those under CN_1 and CN_3. All the upper-layer temperatures under CN_3 exceeded 60 °C at 48 h and reached the maximal 62.2, 60.9, and 61.1 °C under R_1, R_2, and R_3, respectively. Under CN_2, the maximal temperatures under R_1, R_2, and R_3 were obtained at upper-layer, middle-layer, and upper-layer positions, respectively, and reached 62.5, 61.9, and 61.5 °C at 72 h. For CN_1, only the upper-layer temperature under CN_1R_1 reached 60.3 °C at 96 h. Therefore, a high C/N ratio of 35 contributed to warming the feedstocks in the first 48 h. However, the aerobic pretreatment under a moderate C/N ratio of 30 also showed a favorable warming trend during the 120-h treatment.

In addition, C is the energy source for microorganisms, while N is the nutrient consumed for cell proliferation. Microorganisms assimilate approximately one part of N for every five parts of C to form cell bodies. Meanwhile, microorganisms generally consume twenty parts of organic C to obtain energy when they assimilate and utilize five parts of C. Consequently, the C/N ratios were decreased after aerobic pretreatment, as shown in Fig. 2 (b). The final C/N ratios under R_1, R_2, and R_3 decreased to 21.16, 22.04, and 20.51 for CN_1, 25.33, 26.40, and 25.94 for CN_2, and 29.70, 30.24, and 30.87 for CN_3, respectively. Therefore, the final C/N ratios of the aerobic-pretreated feedstocks were directly suitable for the following co-AD. In addition, the microorganisms degrading lignocellulose are mainly the actinomycetes and fungi that prefer high temperatures (Lin et al., 2018). However, the lignin removals under different C/N ratios and aeration rates only reached 0.87%–1.18%, and CN_2R_3 was optimum. Therefore, the lignin in the corn stover could not be well removed during the aerobic pretreatment with high solids loading and short duration, which was consistent with the general results of aerobic pretreatment at high solids loading (Wu et al., 2022). Considering the characteristics of temperature rise and lignin removal, a C/N ratio of 30 was favorable.

3.2. Effects of oxygen concentration, aeration rate, and time on the temperature rise and physicochemical properties of corn stover and swine manure during aerobic pretreatment

3.2.1. Temperature rise and moisture content

As shown in Fig. 3 (a), the temperatures rapidly increased and then slightly decreased, following similar fluctuations as in Fig. 2 (a). The upper-layer and middle-layer temperatures were holistically higher than the lower-layer temperatures, and the flank temperatures were close to the middle-layer temperatures. The heating rates under Aer_1 and Aer_2 oxygen concentrations were relatively faster than Aer_3, and the optimal temperature rise occurred under Aer_2. The maximal temperatures of $Aer_2R_2T_3$ at four temperature measuring points exceeded 60 °C at 72–96 h. Only the upper-layer and middle-layer temperatures outstripped 60 °C for $Aer_2R_1T_2$ at 72–84 h, whereas its upper, middle, and flank measurement points simultaneously reached their maximal temperatures at 72 h. The temperature of 65.6 °C obtained at the upper-layer position of $Aer_2R_1T_2$ when 72 h was the highest during the aerobic pretreatment in this study. Therefore, the oxygen concentration and aeration rate interactively affected the temperature rise, and excessive aeration was not conducive to heating up. Aer_2, R_1, and R_2 were favorable for the temperature rise of aerobic pretreatment. In addition, the initial C/N ratio and oxygen concentration range under CN_2 and Aer_2 were identical, whereas the ambient temperatures under CN_2 showed a decreased tendency after 24 h, which was subject to overcast skies, rain, cold fronts, and other stochastic weather influences. In comparison, the ambient temperatures under Aer_2 tended to be warmer from 24 to 72 h, followed by a decrease after 72 h, but still were higher than those under CN_2. Consequently, the temperatures at different temperature measurement points obtained under CN_2 after 24 h were lower than those obtained under Aer_2, and the maximal temperature (62.5 °C) obtained under CN_2R_1 was also inferior to that (65.6 °C) obtained under $Aer_2R_1T_2$ (Figs. 2–3), which indicated that the shock of cold air diminished the activity of microorganisms and weakened the temperature-rise effectiveness of aerobic pretreatment because the aerobic pretreatment experiments were using unheated ambient air for aeration at room temperature.

The heat released during microorganisms breaking down organic matter could evaporate moisture apart from heating the feedstocks. Meanwhile, the leachate continually outflew from the bottom of the reactor. Therefore, an entire downward trend was in MC except $Aer_1R_1T_1$, $Aer_1R_2T_2$, $Aer_1R_3T_3$, $Aer_2R_2T_3$, and $Aer_3R_1T_3$ showed a slight upward trend in the first 24 h, as shown in Fig. 3 (b). A possible reason for the increased phenomenon is that the moisture generated by microbial degradation of organic matter was more than that lost during ventilation in the early stages of aerobic treatment (Chowdhury et al., 2014). The MCs under Aer_3 decreased the fastest apart from $Aer_3R_1T_3$, followed by Aer_2, and the moisture lost the slowest under Aer_1. Therefore, the higher the oxygen concentration, the greater the moisture loss. The oxygen concentrations of 17%–20% under Aer_3 were closer to the oxygen concentration of about 21% in the ambient air, and the aeration started as soon as the oxygen concentration dropped to below 17%. The microorganisms could obtain sufficient oxygen, whereas moisture and heat were more lost because the convective latent heat and moisture losses carried away by air flowing through the feedstocks were the main heat losses during ventilation (Ajmal et al., 2020), which was also the key reason for the relatively low temperatures under Aer_3. Although the microbial activity and heat production theoretically were lower under Aer_1 due to less oxygen supply, the moisture and heat losses under Aer_1 simultaneously were lower. Therefore, the actual temperatures under Aer_1 were higher than those under Aer_3 apart from upper-layer temperatures because the temperature rise synthetically reflected the production and loss of heat and moisture in the aerobic pretreatment process.

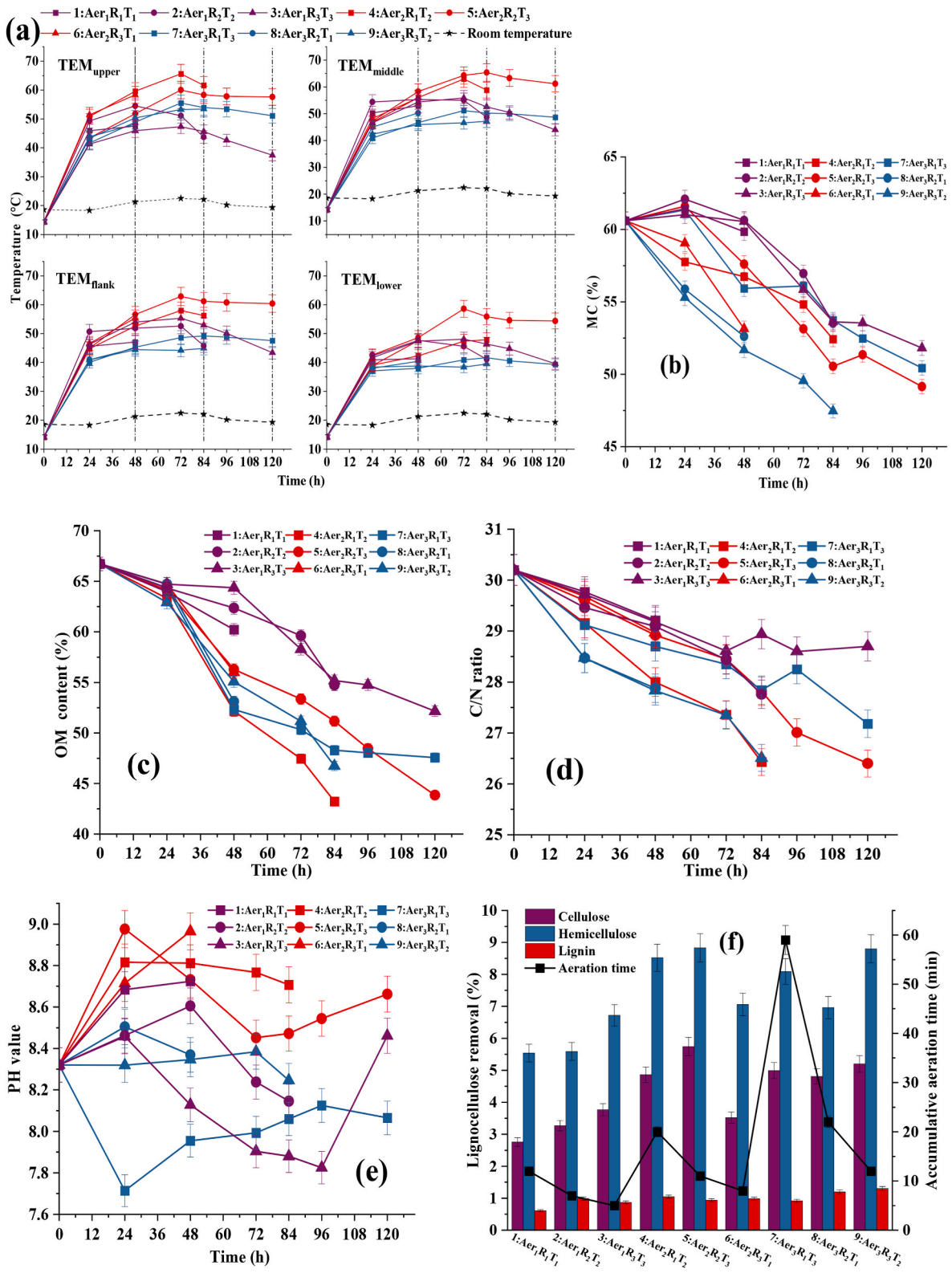


Fig. 3. The effects of oxygen concentration, aeration rate, and pretreatment time on physicochemical characteristics of corn stover and swine manure during aerobic pretreatment. (a) Temperature. (b) MC. (c) OM content. (d) C/N ratio. (e) PH value. (f) Lignocellulose removal and accumulative aeration time.

3.2.2. Organic matter content, carbon-to-nitrogen ratio, pH, and lignocellulose removal

The OM decomposition efficiency characterized the level of microbial activity during aerobic pretreatment and is a comprehensive result of multiple regulatory parameters, such as temperature, oxygen supply, etc. As shown in Fig. 3 (c), the OM content followed a similar decrease tendency to MC. The OM removals under Aer_1 were lower than those under Aer_2 and Aer_3 from 48 h to 120 h. The maximal OM degradation rate was obtained under $Aer_2R_1T_2$ when pretreating for 84 h, with a reduction of 35.22%, which indicated that the ventilation conditions

under $Aer_2R_1T_2$ were most suitable for enhancing the activity of microorganisms. The OM removal of 34.29% under $Aer_2R_2T_3$ when 120 h was higher than those (21.83% and 28.73%) under $Aer_1R_3T_3$ and $Aer_3R_1T_3$. Therefore, the oxygen concentration range of 13%–17% under Aer_2 was favorable for OM degradation.

In addition, an entirely decreased trend was observed in the C/N ratios except for the slight increase under $Aer_1R_3T_3$ and $Aer_3R_1T_3$ at 84 and 96 h, respectively [Fig. 3 (d)], possibly because of higher nitrogen retention than nitrogen degradation. In the first 84 h, the C/N ratios under $Aer_2R_1T_2$, $Aer_3R_2T_1$, and $Aer_3R_3T_2$ descended the fastest. At the end of aerobic pretreatment for 120 h, the maximal C/N ratio decrease was obtained under $Aer_2R_2T_3$, followed by $Aer_3R_1T_3$, and the minimal descent occurred under $Aer_1R_3T_3$. All the final C/N ratios ranged from 26.40 to 29.21 after aerobic pretreatment. Under the same initial C/N ratio of 30 and oxygen concentration range of 13%–17%, the final C/N ratios (26.43 and 26.40) of $Aer_2R_1T_2$ and $Aer_2R_2T_3$ were almost equal to that (26.40) of CN_2R_2, which indicated that the higher temperature rise obtained under $Aer_2R_1T_2$ contributed to the reduction of the C/N ratio and a low aeration rate was favorable. In conclusion, the final C/N ratios of the feedstocks undergoing aerobic pretreatment under dynamic aeration could meet the C/N ratios required for the following co-AD.

The pH values increased in the first 24 h except for $Aer_3R_1T_3$ and $Aer_3R_3T_2$ [Fig. 3 (e)]. Although most reports elucidated that the pH value slightly decreased in the early stage of aerobic treatment because microorganisms rapidly utilized the dissolved OM to produce small molecular organic acids, some studies also found that the pH value might be unvaried or slightly elevated because the feedstocks, for example, manure, biogas residue, etc. or the weak alkaline ammonia generated during ammoniation process of organic nitrogen acted as a buffer (Rodrigues et al., 2021). Furthermore, Beck-Friis et al. (2001) found that the temperature shift from the mesophilic phase to the thermophilic phase occurred in parallel with the pH shift from acidic to alkaline in the initial stage of aerobic treatment and bacteria with a high rate of organic decomposition under alkaline circumstances proliferated as pH increased, speeding up the elimination of OM. Therefore, the pH values showed different trends of increasing and then decreasing, or decreasing and then increasing after 24 h with the mineralization and utilization of organic acids by microorganisms, nitrification and conversion of ammonium nitrogen into nitrate nitrogen, volatilization of free ammonia during aeration, and continuous decomposition of OM and organic nitrogen (Zhu et al., 2021), reflecting the dynamic equilibrium of VFAs from OM decomposition and ammonia from organic nitrogen ammonification. The pH values ranged from 7.71 to 8.98 during the aerobic pretreatment in this study, which was similar to the optimal pretreatment range of pH (7.96–8.80) obtained by Chen et al. (2013) for VFAs production during the AD of waste-activated sludge and food waste.

The removals of lignocellulose containing cellulose, hemicellulose, and lignin increased as the pretreatment duration increased, as shown in Fig. 3 (f). The lignocellulose elimination efficiencies under Aer_1 were relatively lower than those under Aer_2 and Aer_3. Although the pretreatment time of 120 h under $Aer_3R_1T_3$ was longer than that under $Aer_3R_3T_2$ (84 h), the lignocellulose removals of $Aer_3R_1T_3$ were not superior to those of $Aer_3R_3T_2$. Furthermore, the hemicellulose removal rates (5.54%–8.83%) were higher than those of cellulose (2.76%–5.74%) and lignin (0.61%–1.30%) because hemicellulose was a heterogeneous polymer composed of monosaccharides including xylose, arabinose, and galactose, and was more unstable than those of the cellulose with macromolecular polysaccharides and the lignin with phenylpropanoid. Lignin removals obtained under $Aer_2R_1T_2$, $Aer_3R_2T_1$, and $Aer_3R_3T_2$ only reached 1.05%, 1.20%, and 1.30%, respectively, and were similar to the lignin removal of 1.18% obtained under CN_2R_3, which further elucidated that the lignin removal capacity of aerobic pretreatment at high solids loading was low. In addition, the oxygen concentration, aeration rate, and aerobic pretreatment time interactively affected the cumulative aeration time, which determined the energy consumption and pretreatment effect of aeration operation during aerobic pretreatment. The accumulative aeration time of 59 min obtained under $Aer_3R_1T_3$ was the longest. The aeration durations were similar for $Aer_2R_1T_2$ and $Aer_3R_2T_1$ (20 and 22 min) but did not exceed 12 min under other pretreatment conditions.

3.3. Effects of aerobic pretreatment under dynamic aeration on co-anaerobic digestion characteristics of corn stover and swine manure

3.3.1. Methanogenesis

As shown in Fig. 4, the accumulative methane production of the control only reached 4.37 L, which corroborated that the dry co-AD of corn stover and swine manure without pretreatment could not run smoothly. The daily methane production of $Aer_2R_1T_2$ increased rapidly on the second day, reached the maximal peak value on the fifth day, then slightly fluctuated up and down within 6–8 d, and peaked again on the eleventh day after a significant fall because of acidification. $Aer_2R_2T_3$ showed two peak values at 12 d and 16–19 d, respectively. Relatively low peaks occurred between 11 and 17 d, respectively, for $Aer_2R_3T_1$, $Aer_3R_1T_3$, and $Aer_3R_2T_1$. Although $Aer_1R_1T_1$ and $Aer_3R_3T_2$ reached their peaks earlier in the first five days, their peak values were lower than $Aer_2R_1T_2$. Therefore, $Aer_2R_1T_2$ outperformed other samples, and its total cumulative and VS methane production for 25 days reached 338.76 L and 258.93 L/kg VS, followed by $Aer_2R_2T_3$ with 255.97 L and 234.86 L/kg VS. Two similar cumulative and VS methane yields (122.72 and 134.34 L and 201.50 and 196.83 L/kg VS) were for $Aer_2R_3T_1$ and $Aer_3R_1T_3$. Approximate yields also occurred for $Aer_3R_2T_1$ and $Aer_3R_3T_2$ (101.80 and 107.87 L and 142.99 and 155.66 L/kg VS). The methane yields under Aer_1 were not up to par and only reached 63.84, 33.66, and 42.71 L and 90.21, 60.04, and 81.68 L/kg VS, respectively, for $Aer_1R_1T_1$, $Aer_1R_2T_2$, and $Aer_1R_3T_3$. Therefore, the aerobic pretreatment under dynamic aeration based on the oxygen concentration range of 13%–17% promoted the rapid start-up of co-AD and increased methane production of corn stover and swine manure, and a lower aeration rate and a moderate pretreatment time were more favorable for the subsequent co-AD. Although aerobic pretreatment with high solids loading had the potential to partially destroy the crystalline structure of lignocellulose and hydrolyze some macromolecular OM into tiny molecules of organic acids that microorganisms could use, microorganisms simultaneously devoured some OM for their metabolism and reproduction, particularly the soluble OM in swine manure, thereby the overall OM amount was decreased for co-AD. Therefore, excessive pretreatment time was not advisable.

3.3.2. PH, ammonia nitrogen, and volatile fatty acids

The variations in pH characterize a synergistic effect of the solvation equilibrium of CO_2 and H_2S between the vapor and liquid phases, the acid-base balance in the liquid phase, and the ion dissolution equilibrium between the solid and liquid phases (Wang et al., 2020). As shown in Fig. 5 (a), The initial pH values of pretreated corn stover and swine manure ranged from 6.71 to 7.35 at the beginning of co-AD and the pH values showed significant downward and then upward fluctuations in the first nineteen days of co-AD, indicating that self-buffering system of CO_3^{2-}/HCO_3^- and $NH_4^+/NH_3{\cdot}H_2O$ had not yet formed (Mao et al., 2017). The pH values gradually stabilized after 19 d and reached 6.82–7.26 by the end of co-AD as acidification and methanation approached equilibrium apart from the control. In comparison, the pH value of the control fluctuated between 5.31 and 5.73 until the end of co-AD after a sharp decline in the first ten days, which further elucidated that excessive VFAs inhibited the initiation of co-AD.

Although ammonia nitrogen is integral for anaerobic microbial synthesis, excess free ammonia transformed from organic nitrogen might harm microorganisms in the ammonium/ammonia balance due to its easy cell membrane crossing (Ruiz-Sánchez et al., 2018). All the ammonia nitrogen contents varied between 50 and 358 mg/L during co-AD, as shown in Fig. 5 (b), which was less than the ammonia nitrogen

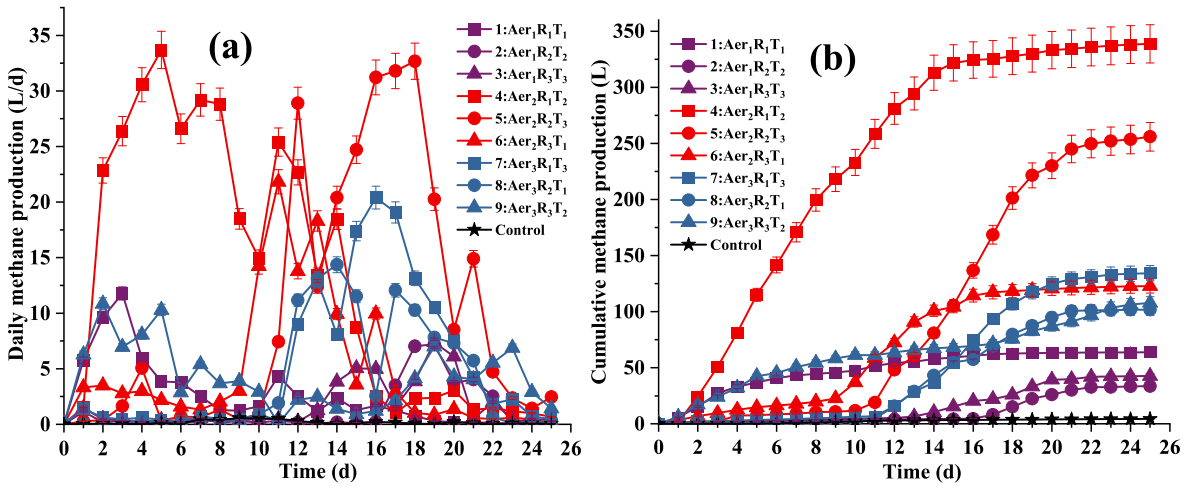


Fig. 4. Methane production of aerobic-pretreated corn stover and swine manure. (a) Daily methane production. (b) Accumulative methane production.

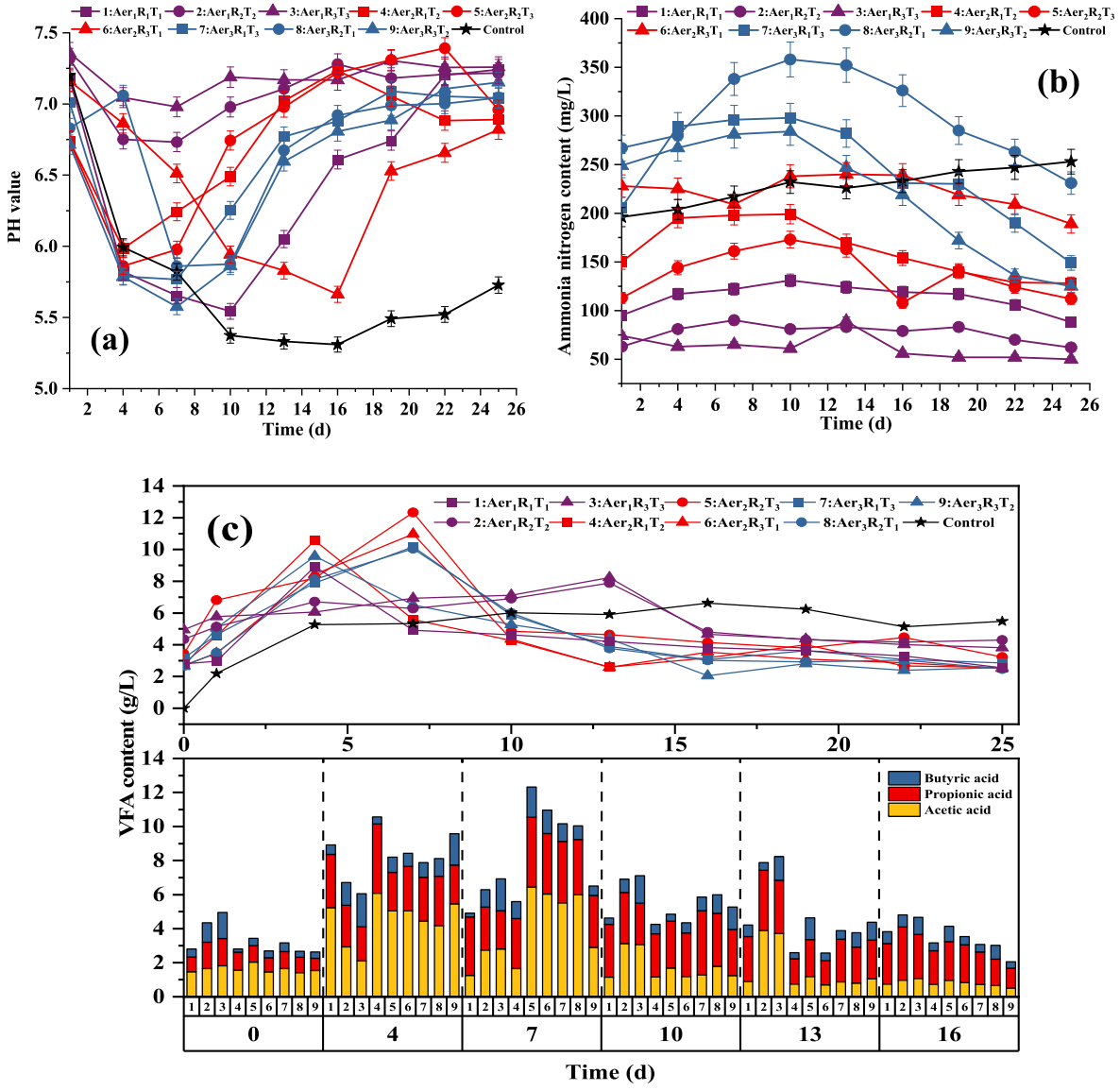


Fig. 5. Variations in pH value, ammonia nitrogen content, and VFA content of leachate during the dry co-AD of corn stover and swine manure. (a) PH value. (b) Ammonia nitrogen content. (c) VFA content.

inhibition limits of 670 mg/L and 2000 mg/L for the production of VFAs and methane due to the acidification functionality of corn stover in the feedstocks in this study (Chen et al., 2021). In a well-functioning AD system, the ammonia nitrogen usually first shows an increasing trend until the ammonium nitrogen consumed by hydrolytic and methanogenic bacteria and the free ammonia lost through evaporation exceed the ammonia nitrogen amount produced from the decomposition of proteins and amino acids, and then begins to decrease and ammonia nitrogen and VFAs maintained a dynamic equilibrium (Rubežius et al., 2020). However, the ammonia nitrogen contents of the control in this study continuously increased during co-AD. The ammonia nitrogen contents were the highest under Aer_3, followed by Aer_2, and the ammonia nitrogen contents were the lowest under Aer_1. Amino acid oxidizing enzymes converted more amino acids into ammonia nitrogen under Aer_3 due to an adequate oxygen supply (Rubežius et al., 2020), thereby maintaining lower C/N ratios and higher ammonia nitrogen contents at the onset of co-AD and even during co-AD. In comparison, the ammonification reactions of Aer_1 were not active during co-AD because of poor pretreatment effect and relatively high C/N ratio. Therefore, the ammonia nitrogen levels fluctuated less under Aer_1.

Methanogens need to decompose VFAs to generate methane, and VFAs are critical intermediates in the AD process (Chu et al., 2021). As shown in Fig. 5 (c), the initial VFA contents (2.63–4.95 g/L) of pretreated corn stover and swine manure were higher than that of the control, which further elucidated that aerobic pretreatment could accumulate a certain amount of VFAs and active hydrolytic acidification microorganisms for the subsequent co-AD, thereby accelerating the start-up of co-AD (Saritpongteeraka et al., 2015). The variations in VFA contents followed the typical pattern that first climbed quickly, then declined, and finally tended to stabilize. The VFA peaks under Aer_2 were relatively high. $Aer_2R_1T_2$ peaked on the fourth day, while the maximal peak value was under $Aer_2R_2T_3$ on the seventh day. The VFA content of the control showed a significant increase in the first four days before becoming essentially constant, which matched its variations in methane production and pH value and further defined that irreparable rancidity occurred. Although the pH value and the ammonia nitrogen and VFA contents of $Aer_1R_2T_2$ and $Aer_1R_3T_3$ did not show inhibition, the methane produced was less throughout co-AD. Therefore, a pseudo-steady state might be exhibited under Aer_1 due to the interactions between pH, ammonia nitrogen, and VFAs, characterizing stable operation but low methane production (Lü et al., 2013). In addition, the VFAs mainly included acetic, propionic, and butyric acids. Acetic acid and propionic acid showed similar fluctuations to the overall VFA level and peaked at 3.72–6.45 g/L and 3.04–4.10 g/L at 4, 7, and 13 d, respectively. Xu et al. (2020) reported that approximately 70% of the methane came from acetic acid during AD. Rubežius et al. (2020) found that aerobic pretreatment promoted the acetate acid conversion from VFAs in the AD process of poultry manure, while acetic acid did not constitute a high portion of the total VFAs after ten days in this study because of the microbial predilection utilization for acetic acid. The propionic acid level at the start of co-AD exceeded the inhibitory threshold of 900 mg/L (Wang et al., 2009) due to the presence of corn stover in the feedstocks. Yu et al. (2018) reported that the more corn stover added, the more propionic acid accumulated during AD. However, propionic acid did not appreciably impede methane production in this study since it could be oxidized into acetic acid when the acetic acid content gradually fell and consumed by methanobacteria (Wang et al., 2022a),.

In conclusion, a moderate oxygen concentration range of 13%–17% (Aer_2) and a lower aeration rate of 0.1 $m^3/(min{\cdot}m^3)$ (R_1) were preferable for the aerobic pretreatment of corn stover and swine manure with high solids loading. $Aer_2R_1T_2$ was optimum for temperature rise and methane production but reached its maximal temperature at 72 h instead of the 84 h corresponding to T_2. Furthermore, the aeration time would extend to achieve the target oxygen concentration at a reduced aeration rate. Meanwhile, aeration time increased with the duration of aerobic pretreatment. Therefore, the pretreatment duration was further studied in the following study to examine the feasibility of a 72-h pretreatment duration.

3.4. Effects of pretreatment duration, inoculum loading, daily leachate reflux repetitions, and inoculation layers on methane production of aerobic-pretreated corn stover and swine manure

The effects of pretreatment durations (72, 84, and 96 h for A_1, A_2, and A_3), inoculum loading (20%, 30%, and 40% for B_1, B_2, and B_3), daily leachate reflux repetitions (once, twice, and thrice for C_1, C_2, and C_3), and inoculation layers (single-layer, double-layer, and triple-layer for D_1, D_2, and D_3) on methane production and VFAs variations are as shown in Fig. 6.

As shown in Fig. 6 (a), although the accumulative methane yield of A_2 was somewhat less than that of $Aer_2R_1T_2$ under the same 84-h pretreatment, the daily and accumulative methane production under A_2 was superior to those under A_1 and A_3. A prolonged pretreatment period of 96 h under A_3 was not favorable. The daily methane production of A_2 peaked in the first five days and was earlier than A_1 and A_3. The cumulative methane yields A_1, A_2, and A_3 were 292.64, 308.05, and 258.16 L, respectively, and their methane contents were 36.98%–42.81% in biogas. The VS methane yields of A_1, A_2, and A_3 reached 229.95, 247.79, and 205.58 L/kg VS, respectively, and the maximum at 84 h was slightly lower than that (258.93 L/kg VS) under $Aer_2R_1T_2$. In addition, increasing the oxygen concentration and aeration rate might increase the power consumption and instead exacerbate the harmful impact of cold air on pretreatment and was not advisable. Therefore, the optimal oxygen concentration range, aeration rate, pretreatment duration of 13%–17% (Aer_2), 0.1 $m^3/(min{\cdot}m^3)$ (R_1), and 84 h (A_2) were used for the following trials to further improve methane production in terms of inoculation method and leachate reflux.

As shown in Fig. 6 (b), the daily and cumulative methane production of B_1, B_2, and B_3 represented a rising tendency with an increase in inoculum loading and peaked within the first five days. The cumulative and VS methane outputs reached 227.73, 343.40, and 444.58 L and 190.25, 249.68, and 266.30 L/kg VS, respectively, for B_1, B_2, and B_3, and the optimal B_3 with inoculum loading of 40% was superior to $Aer_2R_1T_2$ and A_2 with inoculum loading of 30%. The average methane contents in biogas reached 44.94% for B_3 and were increased by 6.93% and 3.24% compared to B_1 and B_2. Therefore, the inoculum loading was crucial for the pretreated corn stover and swine manure to enter co-AD rapidly and generate methane efficiently because the inoculum riched in hydrolytic acidification and methanogenic microorganisms (Yue et al., 2023).

As shown in Fig. 6 (c), the daily methane production of C_1, C_2, and C_3 peaked in the first five days. The cumulative and VS methane yields reached 271.40, 337.02, and 404.42 L and 219.33, 234.74, and 260.14 L/kg VS, respectively, for C_1, C_2, and C_3, and the leachate reflux repeated three times per day was optimum. Meanwhile, the methane contents of C_3 increased by 4.37%–7.91% compared to C_1 and C_2. Multiple leachate backflows were beneficial for promoting methane production in the dry co-AD process because the leachate with higher moisture content could not only dilute the inhibitory compounds affecting AD and enhance the heat and mass transfer characteristics of the feedstocks but also improve the opportunities for microorganisms to contact with the substrates and promote the degradation of organic compounds (Song et al., 2023). Although methane bacteria are sensitive to leachate flushing because the methanogenic bacteria typically require a longer time of 10–15 d to reproduce compared to the 24–36 h and 80–90 h of the hydrolysis and acidification microorganisms (Deublein and Steinhauser, 2008), the leachate reflux of thrice was still favorable in this study. In addition, the methane yield of C_2 was lower than that of B_3 at the same inoculum loading of 40% and daily leachate reflux repetitions of twice, and the methane production of C_3 still did not exceed that of B_3 even when leachate reflux was increased to thrice every day. There are two possible reasons for this phenomenon: (1) the cold air weakened the effectiveness of aerobic pretreatment; (2) the

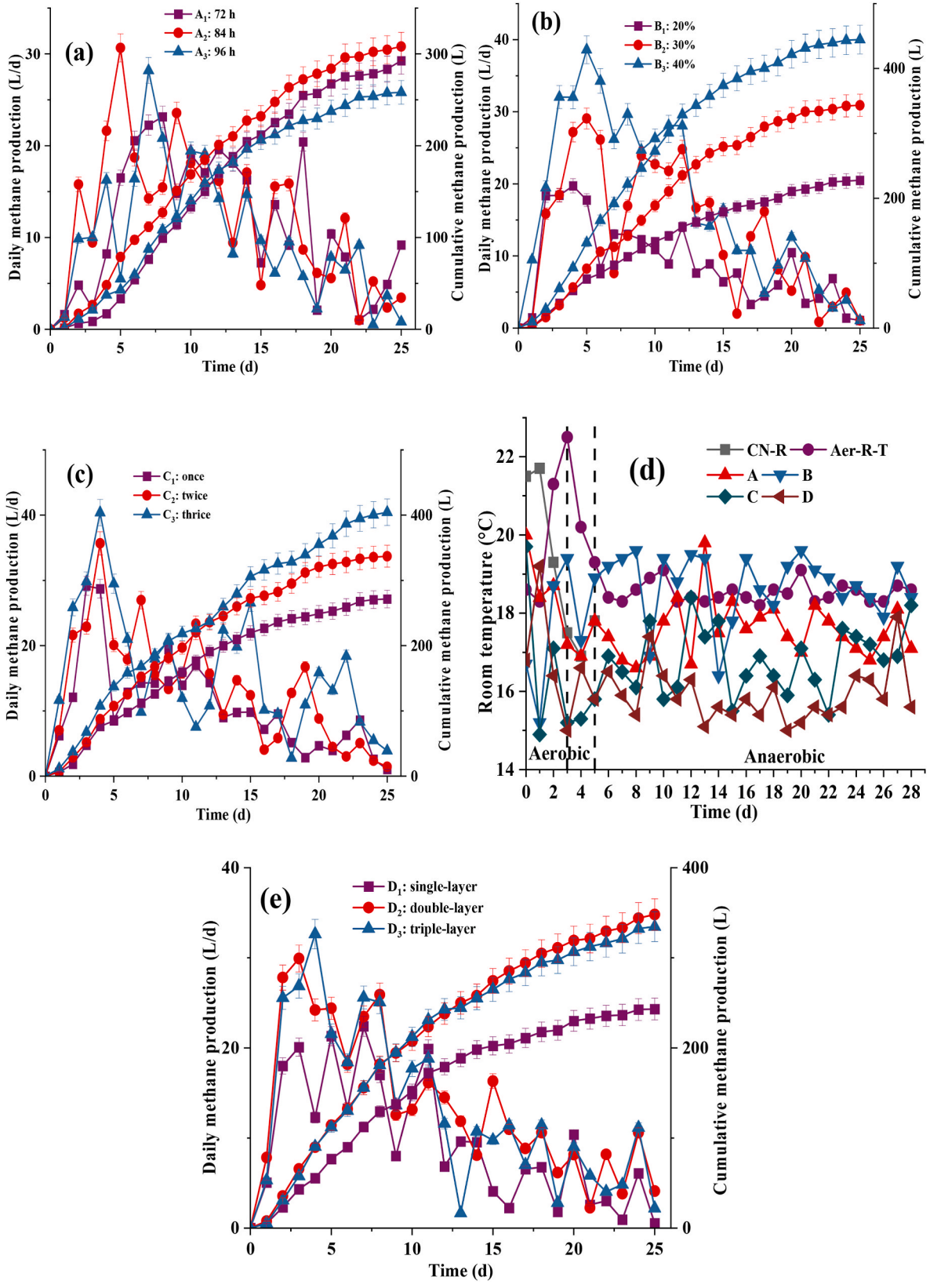


Fig. 6. Effects of pretreatment time, inoculum loading, leachate reflux repetitions, and inoculation layers on methane production of the aerobic-pretreated corn stover and swine manure. (a)–(c) and (e) Methane production under different pretreatment times, inoculum loading, leachate reflux repetitions, and inoculation layers. (d) Room temperature. (e) VFA content.

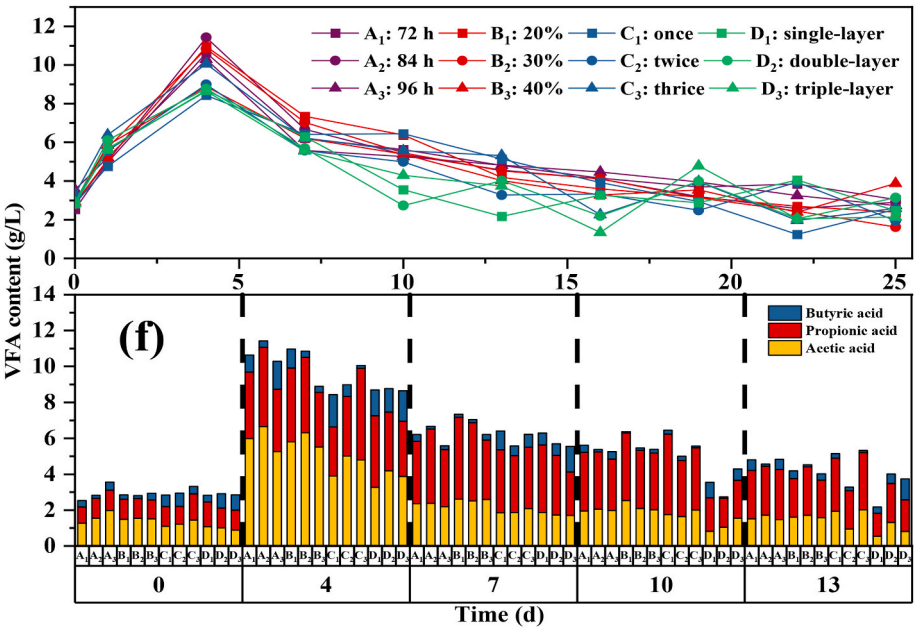


Fig. 6. (*continued*).

low-temperature leachate, which had not undergone secondary heating, had an adverse influence on the co-AD. Based on the results obtained from sections 3.1 and 3.2.1, the cold air affected the temperature rise in the aerobic pretreatment. The ambient temperatures were further studied to elucidate the two reasons, as shown in Fig. 6 (d). The pretreatment, inoculum, and leachate reflux conditions of $Aer_2R_1T_2$, A_2, and B_2 were uniform, while the ambient temperatures of $Aer_2R_1T_2$ were the highest in the first five days incorporating aerobic pretreatment of 3.5 days and co-AD of 1.5 days. The ambient temperatures of B_2 fluctuated less throughout the aerobic pretreatment and co-AD processes. The ambient temperatures of $Aer_2R_1T_2$ at 6–22 d were lower than those of B_2, while their ambient temperatures were similar after 22 d. The ambient temperatures of A_2 were minimal during the aerobic-anaerobic digestion process. Therefore, the accumulative methane production (343.40 L) of B_2 was the highest, followed by $Aer_2R_1T_2$ (338.76), and the 308.05 L of A_2 was the least. The methane production (444.58 L) of B_3 could be increased significantly through the increase of inoculum amount from 30% to 40% compared to $Aer_2R_1T_2$, A_2, and B_2. However, the ambient temperatures of C_2 were lower in the overall aerobic pretreatment and co-AD process than those of B_3, leading to a reduced methane production for C_2 compared to B_3. Therefore, the optimization of dynamic aeration, inoculum addition, and leachate reflux improved the aerobic-anaerobic digestion system's ability to resist low temperatures to some extent and reduced the effects of low ambient temperatures on temperature rise and methane production of corn stover and swine manure.

As shown in Fig. 6 (e), the daily and cumulative methane yields of D_2 and D_3 were higher than those of D_1. Therefore, inadequate contact between the substrates and microorganisms occurred under D_1, and inoculating with three layers was unnecessary. The cumulative and VS methane yields were 243.07, 347.97, 334.56 L and 226.95, 259.16, and 245.53 L/kg VS, respectively, for D_1, D_2, and D_3. Meanwhile, the methane contents of D_2 and D_3 were also higher than those of D_1 by 4.34%–11.92%. In addition, the VFA concentrations rose rapidly, peaked on the 4th day, and then sharply fell to a steady level, as shown in Fig. 6 (f). The primary constituents of the VFAs remained to be acetic, propionic, and butyric acids, and the acetic and propionic acids exhibited a similar tendency to the VFA content. Furthermore, the ambient temperatures of D_2 were close to those of C_3 in the first 11 days but lower than those of C_3 from day 12 [Fig. 6 (d)]. Therefore, although the pretreatment, inoculum, and leachate flux conditions of D_2 and C_3 were identical, the methane production (347.97 L) of D_2 was lower than that (404.42 L) of C_3, which further elucidated the effect of low-temperature leachate on co-AD.

In conclusion, the inoculum loading, leachate reflux repetitions, and stratified inoculation were critical to ensuring the well-functioning co-AD of aerobic-pretreated corn stover and swine manure. Although the decrease in ambient temperature diminished methane production, the methane yields achieved under the optimal conditions of inoculum loading of 40%, daily leachate reflux repetitions of thrice, and double-layer inoculation were still favorable, and the inoculum loading was more correlated with methane production than leachate reflux repetitions and stratified inoculation. The VS methane yields of $Aer_2R_1T_2$, A_2, B_3, C_3, and D_2 reached 258.93, 247.79, 266.30, 260.14, and 259.16 L/kg VS, respectively, and were similar to and even exceeded the methane yields (0.12–0.241 m^3/kg VS) obtained by some studies about dry AD of lignocellulosic biomass and livestock manure (Idefonso et al., 2020). Therefore, the aerobic pretreatment with high solids loading under dynamic aeration was favorable for dry co-AD of corn stover and swine manure, and integrated aerobic-anaerobic digestion was a competitive approach for the low-cost operation of dry AD at different scales.

4. Conclusions

Integrated aerobic-anaerobic digestion of highly solids-loaded corn stover and swine manure was viable. Dynamic aeration accelerated the temperature rise of aerobic pretreatment and enhanced the methanogenic performance during the dry co-AD of corn stover and swine manure. Inoculum addition and leachate reflux were critical for the dry co-AD process to operate steadily apart from aerobic pretreatment. The optimized regulation of dynamic aeration, inoculum addition, and leachate reflux improved the integrated aerobic-anaerobic digestion system's ability to resist low temperatures to some extent and the inoculum loading was more correlated with increasing methane production than leachate reflux repetitions and stratified inoculation. The optimal conditions obtained in this study were an oxygen concentration range of 13%–17%, an aeration rate of 0.10 m^3/(min·m^3), a pretreatment time of 84 h, and an inoculation loading of 40%, refluxing thrice per day, and double-layer inoculation. The maximal cumulative and VS methane yields reached 444.58 L and 266.30 L/kg VS, respectively.

CRediT authorship contribution statement

Lili Wang: Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Zicong Wang:** Writing – original draft, Investigation, Formal analysis, Data curation. **Zhongjiang Wang:** Writing – review & editing, Project administration, Funding acquisition. **Jingke Zheng:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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